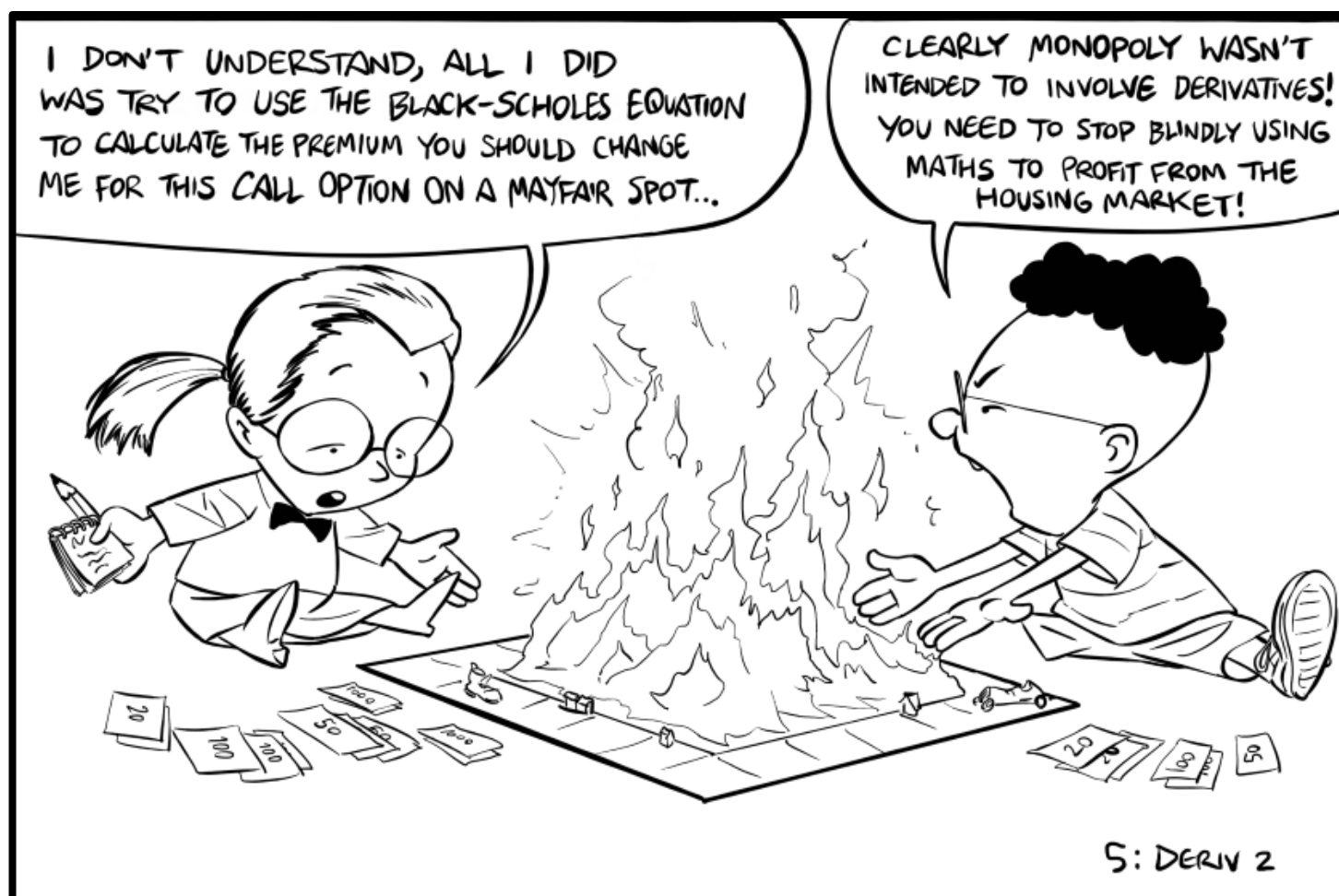


Securities Education Certificate



Exclusive SEC Cartoon by Nicolas Geiseler Toran (Mechanical Engineering, 2022) [5/7]

Lecture 5

Derivatives II

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Options

So, let's summarise futures and forwards. Both involve an agreement between the buyer and seller to buy or sell an asset at an agreed price in the future. The method of settlement and standardised terms of the contract differ between the two, but generally the concept is clear – you agree to buy an asset at some point in the future. One thing to note here: with futures and forwards, you are obligated to complete the transaction – you don't have the *option* to choose whether or not buy or sell the asset at the expiry date (or 'expiration', whichever sounds better to you). Some of you might see where this is going, but if you're still not seeing where this is headed, consider asking yourself:

What if I want the *option* to exercise the contract at expiration?

Well, you'd be in luck – that's where options come in!

Starting with the formal definition, an **option** is a derivative i.e. a contract that offers the buyer (of the derivative) the right *but not the obligation* to buy or sell the underlying asset at a certain date in the future (expiry date), at a specified price (the strike price). More informally, if you buy an option, you get to choose whether you want to buy or sell the asset rather than being forced to buy it at expiration i.e. you get the option to buy or sell (how conveniently named, eh?).

If you've read the section on hedging risks, you might realise that options let you walk away from a deal if you feel like you changed your mind or if the market experiences a huge downturn such that prices are either too cheap to make a profitable sell or too expensive to make a worthwhile investment. Given the greater variability in possible outcomes, options do have more 'strings attached' by having more features associated with the way they work. We discuss here the most basic types individually first but we will later build upon the foundational knowledge of options to see how they work mathematically and how they can be combined in complex ways later to benefit us if the market goes up or down.

Call options

First, let's list out the two main types of options available – call options and put options. Let's start with **call options**. Call options are simple – they are contracts that offer the buyer the right *but not the obligation* to buy the underlying asset at the expiry date, at a predefined strike price. One way to best think about a call option is to imagine picking up the phone to call your stock broker to buy the stock that you're interested in. You would lock in a strike price today and then be able to purchase the underlying asset in the future at that strike price, irrespective of the market price. Ideally, you would want the UL's market price to go up so that you could choose to exercise the option i.e. buy the asset at a cheaper price and then sell it onwards to pocket the difference. However (and this is where you should pay attention), if the UL's market price goes down, your strike price would be more expensive than the market price and you wouldn't want to buy the option's underlying asset at the strike price, since it's cheaper to go into the market and pay less. *As a result, you do not choose to exercise the option and only lose the amount you pay to secure the option in the first place.*

I want to give you an example using a different currency since we've been playing with USD the whole time. There are more currencies than the GBP and USD, so let's try something a bit less vanilla. Consider this example:

I've loved Nissan cars as a kid, but recently I heard that they're going to replace the acclaimed GT-R series with a new model called the Super GT. They've had to stop selling the GT-R due to noise regulations in its

European export markets but I think that the next model is going to be very popular. Nissan cars are reliable (they're Japanese), more affordable and have one of the best brand reputations in the world. Currently, Nissan Motor Co. Ltd trades on the Tokyo exchange (7201:TYO) at 515.50 Japanese Yen (JPY, ¥). I think the price of the stock is going to hit ¥550 this time next year (24/07/23) but I want to buy it this time next year around the current price of ¥515.50 (NOTE – this is just an example and not actually my OWN opinion.)

Right, that's a mouthful there for an example. So, what I want to do is buy Nissan stock in 12 months time at today's price of ¥515.50. What I want to do here is buy a call option that expires on the 24th of July 2023 at around ¥515.50. If I look at an options listing online, I might find an options call being traded at ¥10 for a strike price of ¥516. So, when the 24th of July comes around next year, if I'm correct in my assessment, the market price of Nissan stock is at ¥550 or higher whilst I can be safe and sound in the knowledge that I can buy the stock cheaper at ¥516.

Now, before we draw out a table and do some calculations, I'm going to be clear on a few things here. Options prices are quoted *per share*. Why is this important? Options contracts are standardised, which means that the contract will sell me a specific amount of the UL – in the case of shares, this is 100 shares that each options contract lets me buy. I will put this in bold text to make this clear: **I DO NOT PAY ¥10 FOR THE PRIVILEGE TO BUY 1 SHARE. I HAVE TO PAY ¥10 FOR EACH UNIT OF THE UL I PAY FOR!** So, I have to pay $¥10 \times 100 = ¥1000$, noting that I'm paying for 100 shares of the underlying stock. To the person who's selling me this option call, this "discount-access" privilege fee I pay is called the **premium** of the option call. The trader who's offered me this contract can charge me ¥10 for every share I wish to buy at a discounted price. Futures and forwards have no costs associated with setting them up, but options do – and they're quoted *per share*! A lot of people forget this when they open up their Trading212 account and realise they have to pay 100x the price they're quoted for because of the nature of a call contract.

Once I've paid the premium for the contract, I then have to pay for the 100 shares at the discounted ¥516 I'm quoted on the options contract. That comes to a total of ¥51,600. In total, I've spent ¥52,600 to buy 100 shares of Nissan (in a year's time) but this is better than waiting a year and having to spend the market price of ¥55,000 ($¥550 \times 100$), so I've saved myself ¥2,400 ($¥55,000 - ¥52,600$) that I can use for something else. One more thing... because this is an options contract, I might change my mind and decide "actually, I don't want to buy the 100 shares of Nissan anymore". Of course, you'd first pay the premium to earn the rights you're entitled to in an options contract, but after that ¥2,400 have been paid for, you can choose not to buy the Nissan stock at ¥55,000 if for instance the price per share goes to ¥450. Why would you exercise a contract if you can just pay for the cheaper stock and admit you got it wrong? In that case, you just take the ¥2,400 yen loss, which is purely the premium you paid to get the contract in the first place.

Put options

On the flip side, what are **put options**? Put and call options are complements i.e. they are the opposites of each other! If you inverse everything you read about call options, you have a put option... how wonderful. For completeness, put options are contracts that offer the buyer (of the put option, remember!) the right *but not the obligation* to sell the underlying asset at the expiry date, at a predefined price. Think of a 'put' as you *putting* a stock on the table for someone to buy off you – that's how it's best to remember the concept. A strike price and expiry date is set for you to sell the underlying asset to a willing customer. Ideally, you would want the UL's market price to go down so that while everyone else around you has to sell for cheaper (otherwise known as a 'discount'), you can sell your asset at the higher, pre-agreed strike price and not make a loss. Of course, if the market price does go up in the end, your strike price ends up being lower than the market price so the customer will exercise the option to buy the underlying asset at a discount to them and at a loss to you. Options really are a two-sided game of chess – one side will always win and the other will always lose so you

really want to be sure that you're well prepared to be on the winning side with detailed analysis and an eye for forecasting future shifts in market price and market opinion.

Consider this example:

I think that because of inflation and increasing interest rates, share prices are going to go down such that 12 months from now, my 100 shares of Microsoft (NASDAQ:MSFT) is going to go down in price. This analysis is also supported by the fact that I think the Xbox Series X and Windows 11 aren't going to be very big sellers since market sentiments suggest the PS5 and the new Mac M2 machines are going to gain market share respectively. Right now (06/07/22, 13:00), MSFT is trading at \$262.86. I think that in a year's time (06/07/23, 13:00), it will be at \$230.00 (NOTE – this is just an example and not actually my OWN opinion.)

I want to sell my 100 shares of MSFT in a year's time at around \$262.86 while everyone else might probably be selling and buying shares at \$230.00 on the market. How do I do this? I go on the exchange and buy myself a *put* options contract for MSFT at \$265.00 that expires in a year's time on 06/07/23. Yahoo tells me I can buy myself a call option for MSFT at \$30.00 per share, which I think is a reasonable price. I buy the call option and pay a \$3,000 premium ($\30.00×100) to the trader offering me the quote and wait a year. After 12 months, the price of MSFT is now \$230.00, but I can sell my shares to the trader at \$265.00 each, which he has to accept since I own the contract. He pays me \$26,500, giving me an overall profit of \$23,500 after premium costs have been deducted. If I'd gone to the exchange to sell my 100 shares of MSFT, I'd have sold it for \$23,000, leaving me \$500 shorter – that's the value of 2 shares of MSFT. Thankfully, my trade paid off.

Two small things about options that should be clarified before we talk about the maths behind them:

- Options are standardised to multiples of 100 shares, as explained above. You know that it means there are rules surrounding how they are traded and the conditions that you have to meet in the contract. Those terms and rules are normally set by the exchange on which the option is traded on.
- Options do have a price attached to them unlike futures and forwards. You first pay to buy an options contract (known as a 'premium') and then you get to have the chance to buy or sell the underlying asset in question – we'll come onto this again with another example later when we discuss the maths behind options contracts.

The mechanics behind options

This section is to do with the maths behind a lot of the contracts that you saw above. A preamble before we start: if you don't like maths, we sympathise! A Civil Engineering student and Chemistry graduate like ourselves don't enjoy maths either but there's a reason for it... maths is one of those subjects that's just terribly explained throughout life and lacks the visual understanding that many of us are used to. We are going to try and do a deep dive and make the maths straightforward but if things don't click for you by that point, we recommend you speak to one of us after the lecture or try going online (somewhere like StackExchange) to ask your questions. We've tried our best here to make it simple although we're sure you'll understand that it is easier said than done to make maths easy for everyone.

I'll be clear here: we will teach you these diagrams in detail and we will help you if you do get stuck – we often find this to be a weakness in students throughout the SEC and as a result, we aim to make sure this is well understood this year. They will appear on the exam and although they aren't easy to wrap your head around them at first, we will do our best to go through them step-by-step, in an ordered and linear fashion.

Right, rant over. Let's crack on.

Intrinsic Value

I do hate this phrase because it has many concepts in many different areas. I would make the effort of unifying it all under one umbrella and describe the general notion, but I think that would be too much of a faff. I'll try describing it after I've given its practical definition in the context of options but for now let's focus just on options for a second.

In the context of call options

We'll first wrap our heads around call options, and then move towards put options later.

Let's try another simple example.

Sony trades at \$90 at the NYSE. I like Sony, so I'd like a call option that expires in 3 months time with a strike price of \$95 (Notice here I haven't included the call's premium). 3 months later, the stock price reaches \$100. I get to pay \$95 for a stock that's worth \$100. Fabulous.

So, hypothetically speaking, if everything in the example scenario works out, I get \$5 free profit! Wow. Why does this matter? I mean, we've been discussing similar examples a few times now.

The value of this call option (to me) is, at most, \$5. I would be prepared to pay up to \$5 as a premium to buy this option contract so I can make some profit from immediately selling the underlying Sony stock. Notice here I say 'up to'. If the options contract is priced at \$5, then I pay a \$5 premium to buy a \$95 share, totalling \$100 per share – that's the same as buying the underlying Sony share itself. If the premium was \$4, I would be paying \$99 in total per share, which I could then sell for \$100 each to make \$1 profit each.

In options trading, the difference between the underlying asset price and the strike price is called the **intrinsic value**. Here, the intrinsic value of the call option was \$5, or in other words as I said... I am prepared to pay anything up to \$5 to buy this call option and still make a profit. Pay attention here – the trader also knows that the intrinsic value i.e. the benefit of this options contract is, at most, \$5. If there was no premium i.e. the call option had no cost, I get full access to the \$5 benefit that it offers. A trader being a money-making person would want to maximise their own gains whilst minimising my profits, so they'll try to set the premium to as close as \$5 as possible whilst still enticing me to buy the option contract they're offering.

Mathematically speaking, we denote the intrinsic value as:

$$IV = S - K$$

where IV is the Intrinsic Value, S is the price of the underlying asset and K is the strike price (notice where you've seen this before? That's right, futures and forwards!)

However, let's go one step further. Remember with the example of Nissan when the share price fell to ¥450? I chose not to exercise the option since the strike price of the call option was far higher than the share price. One way to think about this scenario is to consider that the call option had no value – of course, I'm not going to exercise it am I? (assuming I don't go on to sell the contract to someone else, but that's a story for another time). So, what we say if the strike price is higher than the underlying price (i.e. $S < K$) is that the intrinsic value of the option is 0.

Mathematically, we now know that the intrinsic value of a call option is either 0 or the value of $S - K$, provided that $S > K$ (i.e. the inequality $S \leq K$ is not true). We want to be able to select either of these two values when

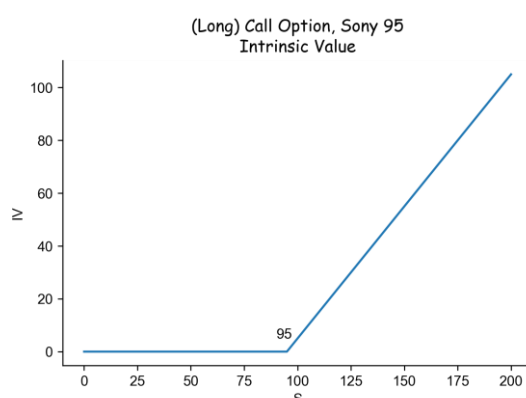
one of these conditions is met, but how do we write it? Maths provides us with a tool called *piecewise functions*.

Piecewise functions

I like to think of piecewise functions the same way I like to think of build-a-bear shops: you can make your own graph the way you want it to behave, especially if nothing in the mathematical world behaves the way you want it to. Let's first write this equation below and then discuss its features:

$$IV = \begin{cases} 0 & \text{if } S - K \leq 0 \\ S - K & \text{otherwise} \end{cases}$$

Now, if we plug our value for K from above (K = 95), we can see our shape on the graph:



How wonderful. For $0 \leq S \leq 95$, the IV of the call option is exactly 0 as invoked by the $S - K$ condition we specified in the domain. Otherwise, if $S > 95$, we find that $S - K > 0$ and we observe a straight line relationship $S - K$, with the gradient equal to 1 implying that the option's IV increases linearly for every unit of currency above our strike price S.

An alternative notation that describes the curve above is the use of the $\max()$ function in mathematics. It's the parent of Python's very own `max()` function, which takes in two values as parameters and selects the value that's bigger. Mathematically, we can write it as such:

$$\max(0, S - K)$$

and test this function at every value of S in the axes above, giving us the exact same graph you saw earlier.

An alternative notation that represents the exact same max function is just to simply put a plus sign in the superscript to denote the positive output of the function:

$$(S - K)^+$$

where the plus (+) simply means that the result of the expression has to be non-negative.

In-the-money, out-of-the-money and at-the-money

Covering the concept of intrinsic value was important for defining **in-the-money** and **out-of-the-money** options. If the intrinsic value for the call option is positive, we say that it is *in-the-money* i.e. the owner of the

option will make a profit if he chooses to exercise his SONY \$95 call option in 3 months if the price of the UL is at \$100. On the other hand, if the price of the Sony stock is at \$90, the call option described in the example above would be *out-of-the-money* as the intrinsic value of the option is 0 i.e. $S - K$ is negative. Interestingly, if the price of the underlying stock was the same as the strike price, $S - K$ would be 0 and we would say the option is **at-the-money**. All of these terms we covered so far are given one specific name... 'moneyness'. Yes, it's an odd name but it's how they're defined in the world of finance.

Notice here that we haven't discussed the option premium. You have to take the premium you pay for the right to an options contract into account, so even if you're at-the-money or in-the-money, you'd have to still account for the option premium before calculating your overall profit so **do not mix up the ideas of in-the-money and profit!**

In the context of put options

Everything you read about call options can be flipped towards put options. They're always complements of one another, which is why learning call options makes learning put options very easy.

Another simple example to kickstart our discussions:

Glencore is facing difficulties in its commodities trading division due to supply chain issues. It currently trades on the LSE: GLEN at £4.00 (or 400.00 GBX). I hold 100 shares in Glencore and I've purchased a put option that expires on the 1st of January 2023 with a strike price of £4.00 and a premium of £0.90. On the 1st of January 2023, Glencore hit rock bottom at £3.00. That's a shame, but I can sell each of my 100 shares at £4.00.

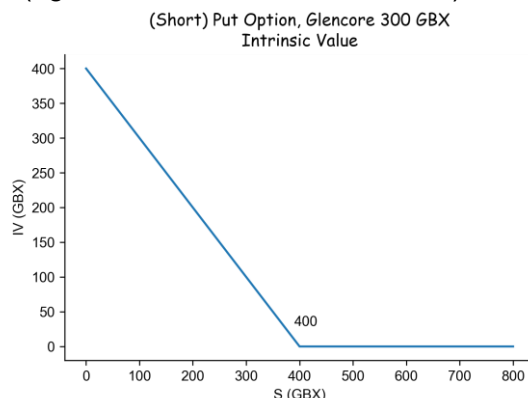
Now, we can speed up our analysis. Notice here that the strike price K is *greater* than our underlying asset price S ($K, S = 4.00, 3.00$). In this case, our formula for intrinsic value is reversed:

$$IV = K - S$$

So, our put option here has an intrinsic value of £1.00. Writing it as a piecewise function is also simple, noting the reverse changes mean that if the price of GLEN goes higher than our value of K at £4.00, our option is worthless and we choose not to exercise it since we could just go on the exchange and sell our shares directly for the higher price.

$$IV = \begin{cases} K - S, & \text{if } S < K \\ 0, & \text{otherwise} \end{cases}$$

giving us a sort-of 'inverted' graph (again, N.B. £4.00 = 400.00 GBX) – see the next page.



So, for $0 \leq S < 4$, the put option contains an intrinsic value and is in-the-money, yet if the price of the underlying stock S is greater than £4.00, the option is rendered worthless, unexercised and hence is out-of-the-money. Notice here the gradient of the line is $-S$, which is significant when you compare it to the scenario for the put-option.

Again, for completion, the notation for a call option can also be written as:

$$\max(0, K - S)$$

or

$$(K - S)^+$$

although I'm pretty sure you're aware that the notation is now self-consistent with that of the call option that we discussed earlier. So, now you've covered the maths and shapes of call **and** put options... congratulations! Let's take this further.

Combined options position strategies

Payoff diagrams

Now things are getting sophisticated! One thing I forgot to mention in the diagrams above is that they're also called **payoff diagrams**. I chose not to mention it at the time since I don't think it captures the true meaning of the graph – the definitions are definitely synonymous (i.e. the same) but the concept of intrinsic value is a better approach to understand the piecewise functions and the associated maths. In essence, from the context of payoff, the graphs show how much money you'll get from the options at certain values of the underlying asset price S .

Another reason that I didn't mention 'payoff' is that thanks to the wonderful world of mathematics, economics and anything numbers-related, if you google 'payoff diagrams' you will get results that are inconsistent and in some cases, completely wrong. How could that be? I don't know, but I certainly feel it's better to be strict on the terminology on a case-by-case basis.

Let's briefly sit in the perspective of payoff when we look at these graphs. If we go back to our put option, we're saying that if the underlying stock is valued at £2, our option 'pays us off' (ha, see what I did there) £2 – we notice that by reading the y-axis or observing the linear 1:1 relationship of the line. However, do we actually pocket £2 for the sale of each share? Do we? No. Remember, we had to pay a premium of £0.90 to be able to buy the option to sell our shares at £4 compared to everyone else who has to sell at the market price of £2. So, how does our so-called 'payoff diagram' actually look when we consider the cost of an option and turn our y-axis into Profit & Loss?

Profit & Loss graphs

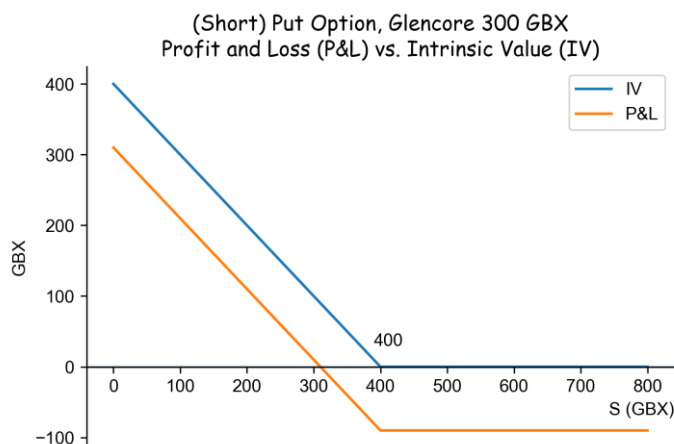
Well, let's think of a simple equation from business studies. We know that profit = revenue – cost. Yes, there's gross profit, net profit etc. but let's just say there's one cost... the cost of an option. Well, we have the functions for revenue... they're the piecewise functions, which we did call 'IV'. Since the cost is a fixed value, we can simply write for the GLEN example:

$$PL = IV - 0.90$$

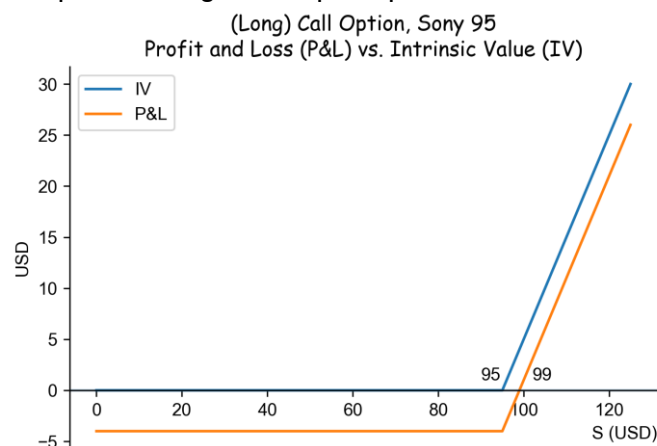
where PL is the profit and loss (P&L) and IV is the intrinsic value, of which we have the required graphs. Some of you might see where I'm headed with this, but if you were in school doing GCSE maths (or something equivalent), you may remember linear translations of a graph in the form:

$$f(x) + C$$

where C is just a constant number (positive or negative) that moves our original graph $f(x)$ up or down respectively. What you can see here is literally the same thing. All we have to do to get a P&L diagram of our put option for GLEN is to translate our IV graph down by £0.90 and we can read our profits at each value of S off the y-axis!



Isn't that magic? The red curve is the curve that represents our profit for the GLEN put option. Let's try the same thing with our SONY call option, taking a call option price of \$4.



The same effect has been achieved. Notice that due to the vertical translation of the graph downwards, our strike price has not moved – it's still in the same place on the x-axis as before. Where our options strategy falls below the y-axis, we would be seeing a loss per share sold if we sold it at that value S . There's a lot that you can gain from profit and loss (P&L) diagrams for payoffs but it's important to understand how they are constructed as you'll find yourself reading inconsistent literature that uses the same words in different contexts.

Short Straddle

We've discussed put and call options separately so far, but they don't have to work in isolation separately. In fact, they are often combined to make new strategies with unique payoff diagrams. The first option strategy we'll be discussing is the idea of a short straddle.

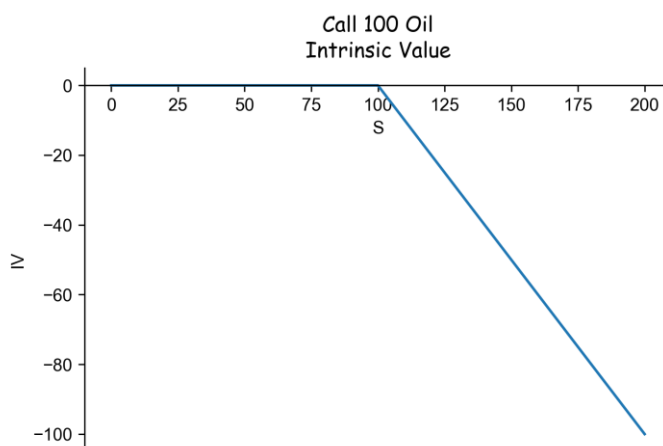
Picture yourself as a trader in a bulge bracket firm. You think prices for a commodity such as oil won't change over the course of the next 3 months. You want to take advantage of this assessment using options to earn a nice premium knowing that the buyers that you're selling this contract to won't be exercising it to purchase the underlying 100 or so barrels of oil.

What you can do here is sell a 3 month call option **AND** a 3 month put option at the exact same time, hoping that the price of oil will stay the exact same. 1 barrel of oil (bbl) trades at \$100 but the call and put options you trade cost a premium of \$0.75, with the underlying size of 100 bbl and strike price of \$100 for both options contracts (remember, you believe the price of a 1 bbl will not change). You collect \$1.50 (\$0.75 × 2) in premiums per bbl, so in total you make \$150 in premiums (the size of the UL is 100 bbl for both contracts). If the price of a barrel of oil is unchanged, congratulations! You made \$150 in premiums and neither of the two counterparties exercised the option.

However, **IF** there is a freak accident and the price of a barrel of oil changes significantly, **ONE** of the counterparties will exercise their option and you will be out of pocket having to buy the 100 barrels and sell it to them at a discount compared to the going market rate. If the price of oil drops to \$90/bbl, the owner of the put exercises their contract and you're forced to buy their 100 barrels of oil at your strike price of \$100. If the price of oil rises to e.g. \$120/bbl, you have to purchase \$12,000 worth of oil and sell it at a discount of \$100 each to the owner of the call option. Bad times indeed.

Still, using the same techniques as we mentioned when we analysed put and call options separately, we can analyse the payoff and the P&L diagrams to see what would happen if the price of the underlying asset S changes. One thing you will realise here is we have to take the view of a trader. Both of the previous call and put options are drawn from the perspective of a buyer e.g. a retail investor. A trader however isn't going to profit from selling a call option with a strike price below the market price. Instead, the trader will be making a loss as the buyer of the option will exercise the contract and the trader will be out of pocket to buy the UL if it's a call option, and vice versa. It's not too hard to think about the payoff diagram of a call option from the perspective of a trader: if the strike price is above the market price ($S < K$), the trader loses no money as the counterparty will not exercise the contract. However, if the strike price is below the market price ($S > K$), the trader will start to lose money. I'm going to keep the notation of 'IV' here although it doesn't really serve a purpose other than consistency with the other diagrams we discussed.

$$IV = \begin{cases} 0, & \text{if } S < K \\ K - S, & \text{otherwise} \end{cases}$$



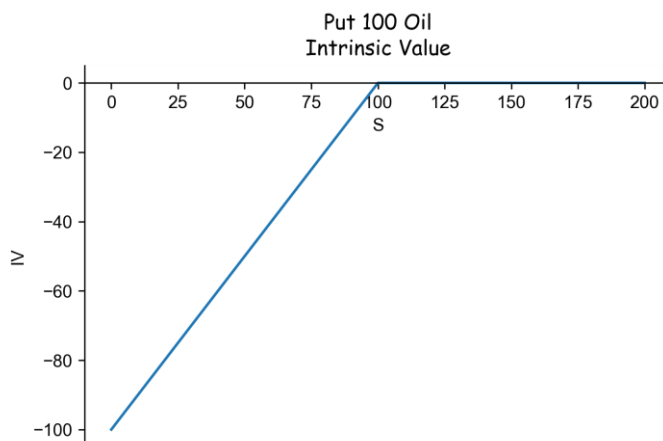
Well, that's interesting. That's the graph we got for the buyer's IV of a call option but it's been flipped on the x-axis i.e. a reflection. Try comparing the two piecewise functions for a trader and a buyer's call option IV and you might see why.

Let's continue this idea for the trader's IV for a put option. If the strike price K is above the underlying price S , the trader would be made to buy 100 barrels of the underlying oil as the buyer of the option exercises the contract. Else, if the strike price K is below the underlying price S , the option isn't exercised by the buyer so the trader doesn't lose any money.

Putting this in a piecewise function gives us:

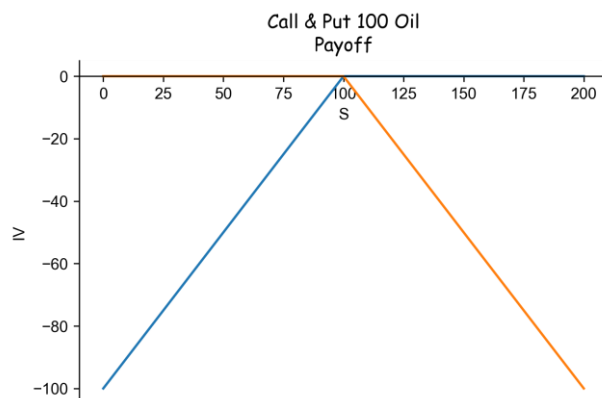
$$IV = \begin{cases} K - S, & \text{if } K > S \\ 0, & \text{otherwise} \end{cases}$$

Plotting this again gives us:

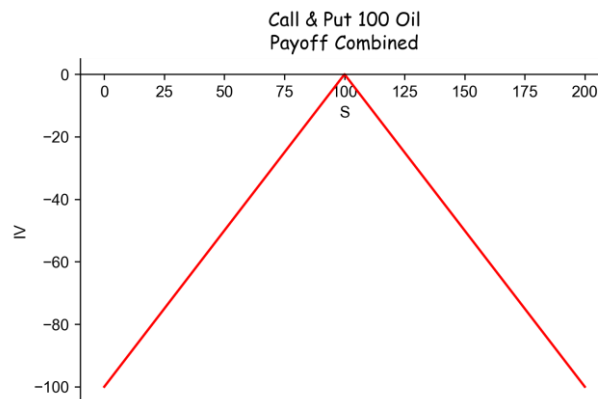


Interesting, the exact reflection of the IV for the buyer of a put option, flipped on the x-axis. Hmm.

If you think about it, selling both a call and a put option means that we can add both graphs together and combine their piecewise functions to view the payoff of the strategy. Let's try that, shall we?



What if we add both graphs together?
Let's try that.



Well well well, we have a triangle of sorts. This here is what we actually call a straddle. Honestly, I have no idea where the name comes from although one definition of it is when you sit on a chair and put one of your legs on either side of the chair feet. Here, it sort of looks like the same idea – on both sides of the strike price K , you have a leg that tails off perpendicular to the other leg.

We haven't considered the cost of the contract. Of course, the trader sells this contract to 2 different counterparties, so we add two units of the cost (which is the total premium to the trader) to get the profit.

$$PL = IV + 1.50$$

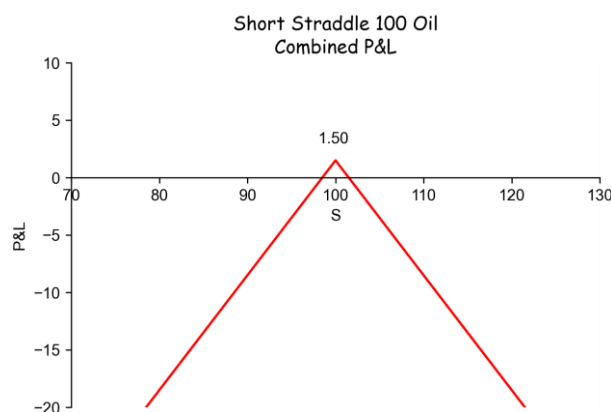
If that doesn't make sense to you, you can also prove the total premium to be 1.50 through some simple algebra:

$$PL_{\text{call}} = IV_{\text{call}} + 0.75$$

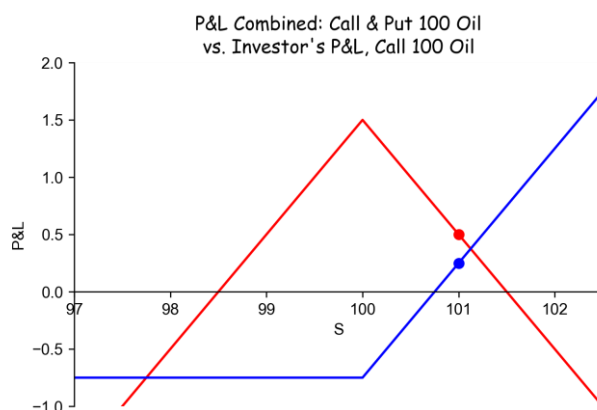
$$PL_{\text{put}} = IV_{\text{put}} + 0.75$$

$$PL_{\text{call+put}} = PL_{\text{call}} + PL_{\text{put}} = IV_{\text{call}} + IV_{\text{put}} + 0.75 + 0.75 = IV_{\text{call}} + IV_{\text{put}} + \mathbf{1.50}$$

Graphing the final result gives us:



A lot of work for one simple graph. You can also see here the sensitive nature of the short straddle. It only takes for the price of oil to change by \$1.50 either way for the trader to start making a loss. Some of you might be wondering “but wait, if the price of the oil is at \$101, that’s higher than the strike of \$100 so how can the trader not be making a loss? The holder of the call option would be exercising the contract, no?” You would be right, as the holder would be netting a profit of \$0.25 per barrel if they choose to exercise their option. However, as the trader owns two units of premium (due to two option contracts being traded), the trader still wins by earning \$0.50 (red dot) in profit per barrel of oil compared to the call holder’s \$0.25 (blue dot).



Later on, you’ll be reading about a very famous case study where a short straddle took down one of Britain’s oldest banks, highlighting the dangers of combining options if you don’t know what you’re doing!

We’ll cover one more options strategy before moving on to other types of derivatives.

A long straddle

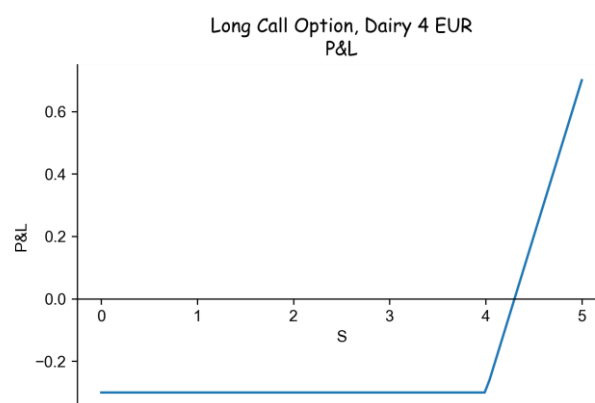
Again, a long straddle is the opposite of a long straddle. I think it’s clear that there’s a pattern now emerging about how long and shorts, calls and puts etc. seem to reflect one another. If some of you are thinking “hmm, what if I flip the short straddle on the x-axis? Does that give me the long straddle P&L shape?”, you’d be right as well! However, we don’t want this lecture to be a simple memory exercise so we’re going to go through the identical analysis (albeit a bit faster) to reach the long straddle shape.

A long straddle involves the simultaneous purchase of a call and a put – that’s basically a long call and a long put combined together. We don’t actually need to reinvent the wheel like we did when we analysed a short call

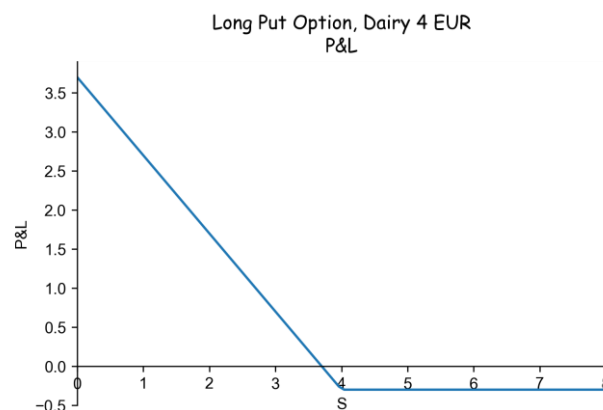
and a short put because we already have gone through the idea of going long on an option very early in this section!

Let's devise another scenario: Imagine you think that dairy is going to swing large in price from €4 per gallon to €6 over the course of a year. You believe this due to the effects of climate change, a change in lifestyle choice and the Dutch choosing to make cheese instead of milk over the course of 2023 (purely hypothetical, of course). Of course, you don't know whether this might swing strongly the other way – who knows, maybe the swiss might repopulate their grassland with cows and increase the supply of milk into the european dairy market. Whatever happens, you want to profit off the large swings in prices compared to the idea of predicting no change in price with a short straddle. You set a strike price of €4 and pay for a call and put option, both at €0.30.

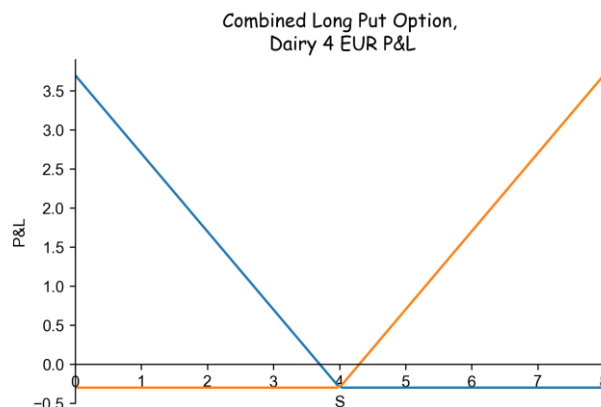
Purchasing the long call gives you the following P&L graph (with the cost accounted for):



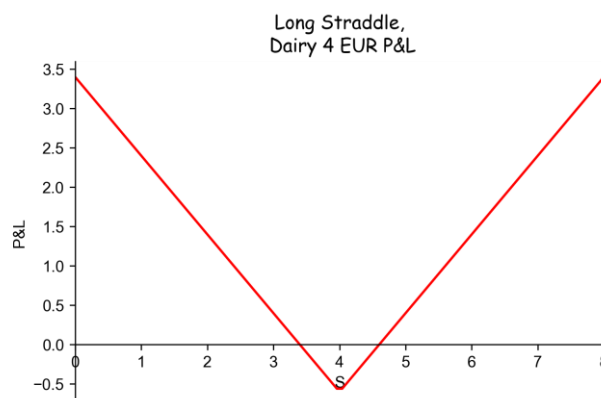
and the long put:



Addition of the two long options gives us the following (see next page):



which gives us (in total):



Wonderful. The long straddle. You can see here as well the potential profits if the price of the underlying milk S goes either side of the $\$4$ to make the trader very rich. What we haven't discussed yet is the potential losses from a short or a long straddle. Let's look at the simpler case first: the long straddle. If you notice, we only go below $y = 0$ at the x -values of $\text{€}3.40$ and $\text{€}4.60$. So, if the price of milk ends up in between either of those values i.e. $3.40 < S < 4.60$, the trader ends up losing money on the strategy. Even then, the most that the trader can lose is the combined premium that he paid for both options contracts. In this case, the trader's losses are theoretically limited, but their gains are unlimited. For all other values of S (even negative ones, as you will discover in lecture 8), the trader makes huge sums of profit per unit of the underlying stock.

However, a short straddle is far riskier. Notice both sides of the short straddle go downwards, down into the negative values of the y -axis. This isn't by coincidence: the most the trader can gain from setting up a short straddle is the combined premium cost of the two options they sold. The losses, however, can be unlimited. In fact, so unlimited that it can take down an entire institution, just like Nick Leeson found out in 1995.

Case study: Nick Leeson

*Before you continue to read this, please note this is **NON-EXAMINABLE**. We won't be asking a single question about this story – this is only for your interests to read about one of the oldest banks in English history to go into liquidation thanks to the trades of one young derivatives trader.*

Nick Leeson was born in February 1967 in Watford, England. He was described as “an average student” at school who found himself working in the futures and options division at Morgan Stanley in 1987, helping the frontline traders at the bank clear successful derivatives transactions.

After being rejected for a promotion as a stock trader at Morgan Stanley, he joined Barings bank two years later in 1989, which was one of England's oldest banks – founded in 1762. Barings was a boutique bank, catering to High Net Worth Individuals (HNWIs) and high value institutions. Being an old bank, it was used to old-style paper-based settlement of transactions in contrast to the modern technological advances in the trading floor and back offices of Morgan Stanley which Leeson was accustomed to. Leeson was responsible for back office accounting where he found the old practices of Barings flawed and easy to exploit.

Impressing his bosses with his confidence and ability to 'get the job done', he was assigned a desk in the Hong Kong office to track down debts of close to £100 million (actually, £200 million when you adjust for inflation). Spurred on by his success to track down the lost debts, Barings offered him a job as a trader in the Singapore's office to be a part of Baring's futures division, trading on the Japanese Nikkei 225 (basically, the FTSE 500 for Japan).

Strangely, Barings bank allowed Leeson to be the head of the bank's Singapore subsidiary whilst also having the ability to access the bank's accounts to his own liking. Leeson's downfall all started when one of his traders mistakenly bought a put option instead of a call option for a client, losing the client £20,000 and putting the bank £20,000 out of pocket in an error account (which Leeson had conveniently numbered #88888 in homage to the lucky number in the eastern markets.) At first, he racked up tens of millions in losses through margin calls to secure his futures trades alongside generating a list of fake clientele to convince auditors his trades were legitimate. Remarkably, he was able to cancel out his losses by July 1993, which only made him try to test his luck more with futures trading.

Believing he was a lucky trader out of his own merit, he then racked up losses of £94 million hidden in his secret error account #88888, growing up to £160 million later on through further futures trading. After securing 75% of the bank's capital at around £350 million to help pay off outstanding fees he owed on his futures, he thought he could again cancel out the debt by betting on the Japanese political and economic landscape to remain stable and unchanged over the course of the coming years. So, he organised a short straddle on the Japanese stock market where he believed that prices wouldn't move and he could use the premiums paid to wind out the debts he'd build up. Unfortunately for Leeson, Japan was hit by a 6.9 magnitude earthquake the very next day, causing the Nikkei exchange prices to spiral downwards and Leeson's short puts were landing him in bigger debts that he had to find money for. The short straddle strategy was risky – so risky in fact that it had only ballooned his losses to £850 million – that's twice the amount of money that Barings bank had in capital. Leeson did try to long futures in an attempt to stem the spiralling debt, believing that the Nikkei exchange would recover from the effects of the earthquake – again, to no avail.

He then fled Singapore the same night after discovering his office had found discrepancies in the accounting books, owing to the secret #88888 that he hadn't declared to auditors now containing an £850 million debt. Barings found itself in immediate liquidation after the Sultan of Brunei pulled out of a deal to rescue the bank alongside a refusal from the Bank of England to bail the bank out of its misfortunes. Leeson was later arrested in Germany and served nearly 5 years in both Germany and Singapore before being released on good behaviour and a cancer scare. Barings, however, was sold to the Dutch investment bank IAG for £1 before being asset-stripped with its various functions sold off in an attempt to cover the losses.

Leeson was then (funnily enough) still allowed to be involved in financial markets, although he later had a film made about him starring Ewan McGregor in 1999. He also became CEO of an Irish football club and even featured on Celebrity Big Brother in 2018, finishing in 4th place whilst bearing witness to the infamous Roxanne Pallett incident that year. During his time in prison, Leeson claimed he'd never heard of Barings bank before applying and revealed that he maintained an intense dislike for his colleagues such that he was only working with them as a "means to an end" and refused to go to after-work drinks with them. Leeson now offers dinner

talks and presentations to companies about risk management, alongside other media appearances.

How mismanagement of funds does pay!

A gentle introduction to the Black-Scholes model

One thing we haven't mentioned yet is how the price of an option is actually calculated. I've been quite arbitrary in choosing an options price that's a bit less than the option's intrinsic value, although that is a predictive measure at best and is never how options are actually calculated in financial markets. No trading floor of a major investment bank is going to be making up option premiums on the spot, are they? Especially if you consider the volume of trades that go through their respective desks every single hour, several hours a day, 5 days a week.

So, you might ask "how are options prices calculated?" Well, here comes the most famous formula in all of financial mathematics... the Black-Scholes formula. It's a Nobel Prize winning formula used to measure the market value of an option leading up to expiration. I'd like to say one or two things about this formula before we discuss it analytically. First of all, it's a very advanced and complex formula to derive so we don't expect anyone to go and read on how it's derived – it's a graduate mathematics topic, so unless you're a mathematics graduate, you won't have come across or be expected to write out its derivation. Second of all, it's a formula. Yes, formulae dictate much of the world around you but the Black-Scholes formula doesn't work all the time for reasons you'll discover in a few moments. We are going to give you a basic background to the formula itself and its associated parameters, but don't feel like it's the holy grail of all mathematical finance. Even Warren Buffet in 1998 said that no matter how well you understand the equation, you still need a back-up grounding on options in the first place since things can go wrong.

Before we discuss the Black-Scholes equation, there are two types of options you need to know – European Options and American Options. They're actually not too dissimilar, apart from one major feature: European options *cannot be exercised until their expiration date*. If the strike price favours the holder of the option before the expiry date, the holder can do nothing to exercise the option to buy or sell the underlying and so they have to wait until the expiration date to do anything about it. However, an American option is more favourable to the holder – they can actually exercise the option before the expiry date.

Think of this simple example: If a call option has a premium of £3 and a strike price of £50, the holder might want to exercise the option if the price of the underlying is £60 a month before expiration. They get a stock that's undervalued by £10, that they can immediately sell on for up to £7 profit (noting the cost of the option i.e. the premium). If the option in question is a European call option, they can't exercise that option until it expires after a month's time, at which point the underlying price could fall to e.g. £52. However, with an American option, they can immediately exercise the option at any time they wish, leaving the holder with an option that they can sell on immediately for up to a £7 profit. Why is this relevant, you might ask? The Black-Scholes option works with the much simpler European-style call option, as you will see in a moment.

Right, here is the equation for all of you that are desperate to find out:

$$C(S, t) = N(d_1)S - N(d_2)Ke^{-rt}$$

C is the price of the European Call option, S is the stock price (i.e. the price of the underlying), K is the strike price, r is the interest rate which we use here as a discount rate and t is the time to expiration.

The formula uses two parameters, d_1 and d_2 :

$$d_1 = \frac{1}{\sigma\sqrt{t}} \left[\ln \left(\frac{S}{K} \right) + t \left(r + \frac{\sigma^2}{2} \right) \right]$$

$$d_2 = \frac{1}{\sigma\sqrt{t}} \left[\ln \left(\frac{S}{K} \right) + t \left(r - \frac{\sigma^2}{2} \right) \right]$$

which it passes through a cumulative distribution function $N(x)$:

$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2}z^2} dz$$

which many of you might realise is the normal distribution function that you covered in your first years of A-level statistics.

Now, that's a lot of terms and equations which I'm aware can look daunting. Let's start with the main equation, $C(s, t)$. Some of you might see that if we hide the $N(d_1)$ and $N(d_2)$ that we have:

$$C = S - K$$

Well, isn't this something we've seen before? Yes. It's the intrinsic value that we discussed above. All that the $N(d_1)$ and $N(d_2)$ terms do is multiply both terms by a number between 0 and 1 (because that's what the normal distribution does) to *weight* the terms appropriately. That's all it is. It's our standard equation to calculate the cost of an option using the intrinsic value but weighted appropriately using a bit more maths to make the calculations more rigid!

There are 5 things that we need to be able to calculate the cost of an options contract, which I've already mentioned in a few cases but I'll repeat again:

1. The stock price, S .
2. The exercise price, K .
 - a. We use both S and K to calculate the intrinsic value, $S - K$.
3. The risk-free interest rate, r . We use this to discount the strike price to its present value today using another term, which I'll mention next.
4. The time to exercise the option, t . Yes, that's what we use. Note that we use the exponential function, $\exp(-rt)$ to discount it. You may need to revise your properties of exponentials to see why the exponential would discount or 'divide' the value of K (hint: it's $-rt$, not $+rt$).
5. The standard deviation of the option's previous returns, σ in log form. This is a new term that we haven't discussed before, but essentially it looks at the previous prices of the stock and works out how much they deviate from the standard mean (hence, the standard deviation). If the price of a stock deviates a lot, the price of the option could end up being much higher than K and hence is more likely to be in-the-money at expiration. If it's a stable-price stock, its standard deviation would be lower and would be less likely to end up jumping to a high price S at expiration.

Now, I mentioned that if S is very high compared to K , the cost of the option C would reflect it by being higher. This is also reflected in the terms d_1 and d_2 . Take a look at the logarithm bracket inside both terms. We see a term $\frac{S_0}{K}$ present inside. If we think about it, if this ratio of S_0 to K is large because the price of the underlying S is so high compared to the fixed options price K , the values of d_1 and d_2 would be higher, and hence the normal distribution function would output a value close to 1, leaving the terms S and K in our original equation largely unchanged. Another way to think about it is that if S is so large compared to K for a *call* option, the holder of

the option would be extremely likely to exercise that option at expiration and hence the cost of the option is priced to take advantage of that certainty since it'll almost certainly happen. If the option was unlikely to be exercised (say, because the strike price and underlying price were so similar), the holder of the call option is unlikely to exercise it as their P&L graph would show a loss and hence the call option would have to be priced much lower in order to reflect its unlikelihood of being exercised and be competitive to be bought.

Of course, time does play a part in the value of an option. The closer the time to expiration for an option, the more worthless the option is i.e. the cheaper it becomes. That might seem counterintuitive to you at first but think about it – if you know that a huge event would have to occur in the next 24 hours to jolt prices up significantly, you would largely know what the price of a stock is going to be tomorrow, right? If you leave a stock at market close at \$50, at market open you'd be very surprised that it jumps straight up to \$60 or \$70 dollars, right? So, there's no point buying an option the night before it's due to expire since the likelihood of a huge price change is rather small. Who knows, there might be a small chance that an earthquake might take place in Tokyo the next morning? This is reflected in the discounting term e^{-rt} term, such that it implies that if t is small, e^{-rt} turns out to be large, scaling up the overall size of the $N(d_2)Ke^{-rt}$ term to make C_0 very small.

The final term to consider is the standard deviation of the option returns, σ . We actually use the log returns to make the data behave better during the modelling process whilst also being easier to deal with when doing calculations. Like we mentioned, if the value of σ is large (implying large standard deviations in the option's prices), we'd expect the cost of the option to be higher than if σ was small. You can actually see that behaviour a bit if we look at d_1 and d_2 . Notice the existence of a $\frac{\sigma^2}{2}$ term in both terms. The fraction containing σ is significant to this behaviour of the option cost. In d_1 , the term exists as $\frac{\sigma^2}{2}$ but in d_2 , it exists as $-\frac{\sigma^2}{2}$. Why does this matter? If σ is large, d_1 gets bigger, returning a value close to 1 for the normal distribution function and hence preserving the value of S . However, for d_2 , the presence of the negative sign means that as σ gets bigger, d_2 gets smaller and the normal distribution function returns a value further away from 1, diminishing the size of the $-K$ term. This means that the calculated option cost is closer to $S - K$ and is therefore more expensive, as we would expect.

The Black-Scholes equation behaves as any maths equation does. We input our values for S , K , r , t and σ to get a value for C . However, what if we had the market price for C , and by doing a bit of googling, we had the values of S , K , r and t to hand. Could we reverse engineer the standard deviation of log returns for a stock call option that we're looking at? Well, yes we can! We can rearrange the Black-Scholes equation and get a value for σ , which we call the **implied volatility**. The reason we call it 'implied' is that it's the value generated by the Black-Scholes equation and it's the value it would use to price that option and output the market price. Of course, it doesn't give you any information about the nature of the future prices that we're predicting – only the standard deviation of them, but traders do like to use it as a guideline for when they're looking at derivatives such as European call options on the derivatives markets around the world.

The Black-Scholes equation also gives us another set of quantities that we can use to assess the behaviour of options, called *the greeks*. We'll cover them later on towards the end of the notes, but for now we'd like you to make sure that you understand the nature of the Black-Scholes equation and all the maths of options that have led us up to this point in the lecture. We also haven't discussed the issues with this equation. First of all, the equation is based on a partial differential equation (PDE). For those of you who have never met PDEs, they're basically a way of expressing how a variable x , y , z etc. are equal to a *rate of change*, which we might denote as dy/dx . In actual fact, PDEs are a lot more complicated than that so I've left the definition of a similar branch of rate of changes called ODEs in the previous sentence as a starting point for you to discover PDEs through some independent research, although it's not the focus of this lecture at all. The 'founders' of the

Black-Scholes equation, Fischer Black and Myron Scholes solved this equation to give the one you saw earlier:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS \frac{\partial C}{\partial S} = rC$$

It comes from an idea in Physics known as Brownian motion, which basically is a branch of physics and maths that tries to predict the random motion of particles e.g. gas particles in a given volume. In maths, this is generalised to an idea known as a Wiener process, named after American mathematician Norbert Wiener in his research of time-dependent random events known as stochastic processes. There are a whole list of issues with the Black-Scholes equation that could take a whole day to list and explain in detail, but we'll list 3 of them here that we feel are the most important to remember:

1. Remember that for r and σ , we had to plug in values to get the value of the call option, C ? We assumed that over the course of the option's lifetime, they won't change. Well, if you're in the real world, you might realise that they most definitely do change. The risk-free interest rate definitely changes, which you'll discover is linked rather indirectly to the interest rate you get from a government bond. The value of σ is pre-calculated before the option is sold, but over time the price of the option will be changing and that will definitely change the value of σ as well. In essence, you have to assume two constants in r and σ which you know won't be constant in the real world, and the Black-Scholes equation fails to account for that.
2. Another one for σ : it's calculated based off the previous prices of the call option it's trying to generate a price for. One thing you have to bear in mind in finance is that *past performance does not indicate future results*. It might work most of the time in fairness, but would anyone have guessed in the run up to January 2021 that GameStop options were going to fly through the roof based on the price of the underlying and their premium costs? The Black-Scholes equation fails to account for that because one of its basic parameters, σ , uses past data as a guideline to predict the current price of the option in the run-up to expiry. If the option price wasn't volatile but you could see demand swinging upwards pretty quickly soon and hence drive up the price, the Black-Scholes equation would be underpricing your option by a lot and leave you open to being arbitrated by your counterparties.
3. The really obvious one – it's a European call option. You can't use the Black-Scholes model for an American call option or even a Bermuda call option at that. It's pretty useless if the world's most complex and most valuable equation can only help you price 1 out of 1000 stocks, although there's definitely other models that try and price them effectively – just do remember that these 'exotic' options with lots more features such as dividends and the ability to exercise early will require much more intricate maths, which can definitely go badly wrong if the model isn't close to perfection.

Right, that's now out of the way. Onto the greeks!

The Greeks

Origins from the Black-Scholes equation

Most of you might remember your A-level studies when you covered the first derivative of a function $f(x)$, which gives the gradient of the tangent at a point x for the function $f(x)$. You then will have covered the second derivative, which shows how fast or slow that gradient changes, often given the term d^2y/dx^2 or $f''(x)$. Further derivatives simply describe the rate of change of the function above it, which seems fairly straightforward for

$d^n y/dx^n$ and $f^{(n)}(x)$ respectively. Those of you who study mechanics are aware of position, the first derivative of speed being the velocity, the second derivative being acceleration followed by jerk, snap, crackle, pop and all the other relevant derivative jargon used to describe a motion of a particle.

The Greeks are principally the analogue of the mechanics derivatives although they're applied to the Black-Scholes equation we studied earlier. First, I'm going to show you the notation and the corresponding quantity in a table. Then, I'll briefly discuss each of the quantities in words to show what they measure and give you a few examples so you're comfortable with them.

Here are the Greeks, along with their derivative notation (otherwise known as Leibniz's form).

Name	Derivative	Symbol
Delta	$\frac{\partial C}{\partial S}$	Δ
Gamma	$\frac{\partial^2 C}{\partial S^2}$	Γ
Theta	$\frac{\partial C}{\partial t}$	Θ
Vega	$\frac{\partial C}{\partial \sigma}$	$\mathbf{V}$
Rho	$\frac{\partial C}{\partial r}$	ρ

Some of you who are new to Imperial might be wondering "what's that funny looking ∂ symbol?" It's a good question, to be fair. That's known as a **partial derivative**, and the symbol is called 'del'. Sadly, I can't explain it in this lecture because otherwise these notes would be 60 pages long but a few videos on the internet from Sal Khan at KhanAcademy would help you understand it better. Just remember that a lot of functions in the real world use several variables e.g. $f(x, y, a, \dots)$ compared to what you know about a single variable $f(x)$, so you would need a way of just getting the derivative for x while leaving everything else unchanged – that's what *partial* derivatives do. They only differentiate with respect to *one part* of the variables involved.

The Greeks are named after a subset of the Greek alphabet. To be honest, we're not quite sure why. Some speculate that it was an idea by Mathematicians to sound noble and prestigious when they discussed finance-related concepts back in the day. Another theory is that it's a nod to the concrete birthplace of finance back in Ancient Greece. Whatever the reason is, we don't know. However, we do expect you to be familiar with each of the Greeks and what they mean, as we'll discuss now.

A simple interpretation of the Greeks

Let's discuss each of the Greeks in a bit of detail.

1. Delta, Δ . The point about delta is it tells you how much the price of the option changes if the price of the underlying changes by \$1. If we're talking about a European call option, a delta has a value between 0.0 and 1.0. So, if we have a stock with an option delta of 0.4, that means that for every \$1 increase in the price of the underlying, our option price increases by \$0.40 **PER SHARE!** Since options contracts are standardised to 100 shares, the total premium would go up by \$40 for every \$1 increase in the price of the underlying. For a European put option, delta would be between 0.0 and -1.0 . If the

underlying price falls by \$1, the option premium decreases by \$40. If our stock moves from \$10 to \$11, our option price might increase from \$1 to \$1.40, for instance.

One way to think about deltas is the price sensitivity of the option to the movements in price of the underlying. A higher Δ means that the price of the option is affected more than if Δ is a lot smaller. For call options, if the option is deep out-of-the-money (OTM), Δ is typically quite low but if the option is in-the-money (ITM), Δ is typically quite high. As a result, people use Δ as a probability gauge to guess if the option is going to land ITM or OTM at expiration. A low $\Delta \sim 0.3$ means it has around a 30% of ending up ITM, but for a $\Delta \sim 0.9$, it has a 90% of ending ITM. However, as the price of the underlying moves, it affects whether the option will finish in-the-money or out-of-the-money. As a result, Δ will change with time, depending on the value of S . To measure the rate of change of Δ , we use Gamma, Γ which is what we discuss next.

- Gamma, Γ . In fact, I've already mentioned gamma just in the previous line. It's basically Delta's best friend. I like to think about it in the same way people think about Kobe and Shaq – both needed each other to work well if you're going to trade options. In principle, Γ measures how much Δ changes if the price of the underlying increases by \$1. If we take our previous example, for our first \$1 increase in the underlying price, our option price C ended up at \$1.40 with a Δ of 0.40 and a Γ of 0.10. If the underlying increases again from \$11 to \$12, Δ first increases by 0.10 to 0.50 (i.e. $0.40 + 0.10$), and then the option price C increases by \$0.50 to \$1.90 ($1.40 + 0.50$).

Now, how does theta, θ come into play?

- Theta, θ is actually considered to be an option's biggest enemy. It's a bit like the effect of Shaq deciding not to show up to training and the LA Lakers as a result losing their play-offs against the Clippers. Essentially, θ is what we consider as the option's **time decay**. Remember, options don't have a fixed value for their price. Options do end up expiring at the expiry date and the closer we get to the expiry date, the more useless the option becomes and hence the lower the option price – we pretty much are near-certain what the price of the underlying is a day before the option expires. Theta basically is a negative decimal that tells you how much the option price decreases every day the owner holds onto it. If we have a θ of 0.2, it means that for every share of the option, the contract loses you \$0.20 a day. For 100 shares, you're losing \$20 every day by holding onto the option. This isn't good if you a buyer of a contract as you might want to sell it on to someone else, but great if you're selling the contract as you know the option is becoming more worthless per day until the value of the option eventually hits 0 and no one is going to exercise it close to expiration provided it's out-of-the-money. One way to think about θ is to think of it like rent when living at home. You have to 'pay' θ to get access to the benefits of Γ and Δ .

Okay, what's next. Ah, yes. Vega.

- Vega, V . Do you remember what σ was all about? If you don't, σ was the standard deviation of the option's price over a given time period e.g. over the last year. We then said that if we didn't know what σ was for a given market price C , we could reverse engineer the Black-Scholes equation to give us a value for σ , which we called **implied** volatility. If you're using a strategy such as a long straddle, you want to bet on an underlying such as a stock with high volatility since if the price swings large either way, you'll profit from the call or put option you bought at the exact same strike price K . As a result, if implied volatility is high, the options price C is generally higher as a result.

Now, how does implied volatility affect the price of our call in a way that we can measure? That's what

vega, V does. V measures how much the price of an option will change if the implied volatility changes by 1% either up or down. Consider our original example for an option with a premium of \$1, an implied volatility of 8% and a value for V at 0.04. If implied volatility goes up to 9%, the premium goes up to \$1.04. If implied volatility goes down to 6%, the option premium goes down to \$0.92 (remembering 2% gives a decrease of \$0.08, twice that for the 1% definition). Vega is also time dependent. Think about it – an option that's got another 40 days to run compared to an option with only 10 days left is going to have more opportunity to experience larger swings in option price than one that's close to expiration.

Vega is useful as well if you are trying to work out what could increase or decrease the option's price over a given time period. Things such as a natural disaster (like we saw in Japan earlier), a general election or a huge company announcement could instantly change the price of the underlying stock, which in turn affects the options price and the implied volatility hugely. As a result, V can shift up or down significantly to reflect this.

Last one coming up... Rho, ρ .

5. Rho, ρ . On reflection, Rho is pretty useless but for some case-specific circumstances. In short, it measures how much an option premium changes if the interest rate increases by 1%. However, interest rates rarely change unless there's something drastic such as inflation on the horizon. Even then, ρ is little affected as the interest rate r is adjusted very rarely for it ever to be of any significance.
EDIT: I wrote these notes in July 2022 and now it's November the 12th, 2022... this didn't age well!

So, there we go. That's all the 5 Greeks that come as a result of the Black-Scholes equation. Greeks help you hedge your options positions and be able to find options that are undervalued yet have the potential to return huge payoffs.

Options and Greeks are related in the same way a football shoe and football technique are related. Learning how a football shoe works in terms of its shape and construction will help you understand where you have to kick a ball to get the most power into the back of the net whereas the technique you develop through practice in training and drills helps you adjust your game so that you can kick a ball from anywhere on the pitch in any given high-pressure situation.

Now onto some more discussions about derivatives before we call it a day!

Objectives of using derivatives

The customisable aspect of derivative contracts allows them to be utilised for a wide range of trader objectives. The strategies used fit into three main types of trading; speculating, hedging and arbitrage. We will now put ourselves in the shoes of each of these types of traders to get a better understanding of how the different derivatives described are really used in industry.

Hedgers use derivatives to reduce the risk faced from potential future movements in the market. For example, this may be done by purchasing forward contracts for a certain commodity that may be required by a company in a year's time. As we discussed, this has the effect of locking in a price for future payments that a company may not want to risk increasing. Similarly, a hedger may use options to provide insurance against risks. Unlike futures and forwards, options contracts offer a way for investors to protect themselves against adverse price movements in the future while still allowing them to benefit from favourable price movements.

Speculators act with opposite motives to hedgers, instead of diminishing market risk, a speculator wishes to take a position in the market betting that the market will either go up or down. Futures and options both provide a way in which a type of leverage can be obtained to magnify a speculator's potential returns. When a speculator uses futures, the potential loss as well as the potential gain is very large. When options are used, the speculator's loss is limited to the premium paid for the options. Some notable derivatives speculators include gamestop retail traders and the legendary Michael Burry (see The Big Short).

We have seen how futures, forwards and options are priced assuming no arbitrage opportunities. This is usually true, but there is in fact a third type of trader with the capabilities to find market inconsistencies and profit from arbitrage opportunities occurring in derivatives markets. Arbitrage involves locking in a riskless profit by simultaneously entering into transactions in two or more markets. Generally there is very little room for small investors to utilise arbitrage opportunities as transaction costs wipe out any profit. Therefore, these traders are only really profitable for large investment banks who can access very low transaction fees.

To wrap up, we will now look at one of the most famous cases of derivatives speculation gone wrong leading to a loss of over €4.9 bn all because of one trader at french bank Societe Generale.

Case Study: SocGen's rogue trader

One of the risks faced by a company that trades derivatives is that an employee who may be responsible for finding arbitrage trades with very low risk may become a speculator in search of large returns to boost their bonus. Jerome Kerviel worked for Societe Generale and in 2005 was promoted to become a junior trader in the bank's Delta One products team. His job was to look for arbitrage opportunities across equity indices. For example if a futures contract on an equity index was trading for a different price on two different exchanges, or was mispriced compared to spot prices as we looked at earlier. An arbitrage may also arise if equity index futures prices were not consistent with the prices of the shares constituting the index.

Kerviel used his knowledge of the bank's procedures to pretend to be arbitraging while in reality he was placing huge bets on his own directional views. He took large positions in equity indices with accompanying fake trades to make it appear as if he was hedged. This position grew over time to over €20 billion.

In January of 2008 his large positions were uncovered by SocGen and were unwound over a 3 day period. This was at the time the biggest loss created by fraudulent activity in the history of finance and cost the bank over €4.9 billion.

Non-examinable material

*The material BELOW is **NOT going to be examined** (hence the italics, bold and underlined red text illustrating the point), so you can relax and read it for your own interests. We thought this would be cool to include for anyone doing interviews or for someone who wanted to get a finer understanding on how derivatives are denoted in the real world. Have a read if you wish and hopefully appreciate the examples we've discussed with regards to derivatives in general.*

History of derivatives

The most interesting aspect of history is that not one historian can ever truly be sure of the first mention of something like derivatives, although the earliest given use is attributed to the use of clay tokens in around 8000 B.C. (around 10,000 years ago, apparently) by the people of Sumer (the earliest known civilisation) with Iraq now in its place – in fact, Iraq seems to be at the centre of much of ancient B.C. history. Back then, when humans first got to grips with the concept of 'money' and a new idea of thinking called 'Mathematics', merchants used clay tokens to give to buyers as a promise that they would deliver a specific quantity of goods at a certain date in the future. These clay tokens would then insert into an *envelope*, much like a children's wooden slot puzzle and have the settlement date baked into the token for confirmation – these are basically believed to be the first forwards contract that existed in pre-Jesus times.

After that, the standardisation of derivatives with specific rules came around the 1700s B.C. in Mesopotamia, where contracts were specifically written out to tie merchants and buyers together on the specific terms of a deal. Back then, the main things that were traded were agricultural goods such as grain harvests and livestock but the main notion was that anything that could be traded between merchants and buyers could have a derivatives contract derived from it to protect against counterparty risk. The first ever recorded derivatives contract actually appears to be written by Thales, a notorious flat-earthier and Greek astronomist who in 500 B.C. successfully negotiated the first ever put options contract for olive oil presses. Although the story has different variations, the most accepted version is that Thales purchased the lease to olive oil presses (basically, just machinery) at low prices and knowing the demand for these presses would skyrocket thanks to him predicting a successful harvest, he could then sub-lease the presses for even higher prices and pocket the difference to make him huge amounts of profit. Some sources claim that Thales predicted (and allegedly influenced) the olive oil prices to also increase in price, furthering his own success by raising the lease prices above normal levels although such sources are few and far between to confirm this narrative.

There are some further developments in the area of derivatives after Jesus was born in 0 B.C. but they aren't really of any interest, although one big use of derivatives was in the Dutch 'tulip mania' era of 1600s, a time when dutch tulip bulbs became so expensive as a symbol of wealth and stature that it is now used to reference any modern-day economic bubble that's imminent to explode after its conception (a bit like NFTs in cryptocurrency, for example). In the 1600s, derivatives contracts to trade tulips were at an all-time high and were actually one of the few ways (allegedly) for anyone to even get their hands on a dutch tulip in the first place. Of course, this only lasted for three years from 1634 such that by 1637 by which point the demand and prices of the tulip crashed completely, although historians often cite the Dutch Stock Exchange in Amsterdam Market as the first ever modern financial stock exchange that behaves identically to those that we have today.

Derivatives continued to exist until their most significant development in the early 70s and 1993. First came the Black-Scholes model, an equation derived from the random behaviour of molecular motion in Physics to calculate prices of derivative contracts. It was one of the biggest contributions that mathematics gave to

economics and finance, earning Myron Scholes a Nobel Prize for Economics in 1997 as they could now write equations for the price of an options based off how much it changes in price over time with relatively successful accuracy, although the equation only really worked when certain conditions (or 'constraints') were applied. Also, since the formula made use of the normal distribution from statistics and probability, the formula was never accurate when compared to the real-world results – it's a bit like modelling the mechanics of a car without air resistance... it works, but it'll never actually behave like the real world that a car is actually driven in.

1993 served a more interesting time in finance as it is often cited as the birth of quantitative finance, an area where advanced mathematics aims to try and find marginal gains in the interest of profit and efficiency. What sparked exponential growth in the area of derivatives was the cancellation of a \$4 bn Superconducting Super Collider in Texas by US Congress under the Clinton administration in October 1993, citing spiralling costs (like most things in debate between academia and politics). As a result, the large number of theoretical and nuclear physicists put out of work were then reportedly assembled by Goldman Sachs to become quants, applying their knowledge of string theory and stochastic brownian motion to generate turnover for the bulge bracket in financial markets such as options trading. Nowadays, derivatives are complex with advanced mathematics dominating the subject in many fronts. Critics such as Nassim Taleb (a famous financial statistician) have argued for the omission of Physicists from financial markets due to their inability to develop accurate models in financial markets where models and theories are rarely replicated in reality, although the philosophical debate of quantitative finance is still up for debate to this day.

Options notation at an exchange

There is a convention to write out an option before searching it on an exchange or online. If we take the Microsoft example from earlier, we can write out the following features:

- Ticker – MSFT
- Year of expiry – 2023
- Month of expiry – 07 (July)
- Date of expiry – 06
- Type of option – Put option
- Strike Price – \$230.00

Notice the underlined characters in the bullet points above. We use this in the the notation used in options trading is to package all of these features above onto one line:

MSFT230706P002300000

Reading this in reverse yields us the same features:

- The first 4 letters are the stock ticker – MSFT
- The next 2 digits are the year of expiry – 23
- The next 2 digits are the month of expiry – 07
- The next 2 digits are the date of expiry – 06
- The single letter denotes the type of option – a Put
- The first 5 digits denote the price of the option to the left of the decimal point i.e. 00230 implies \$230
- The final 4 digits denote the price of the option to the right of the decimal point i.e. 0000 implies .0000

Swaps

A swap is an agreement between two parties (e.g. Two companies) **to exchange cash flows** (e.g. loan interest or coupons on a bond) **on different bases** (e.g. a fixed interest rate vs a LIBOR calculated rate) **on particular dates in the future.**

If this doesn't make sense at first, don't worry, we will quickly get into an example which should make the mechanics clear.

In this definition, a cash flow is any movement of money into or from someone's account such as interest payments and 'different bases' means that the cash flows are calculated on different underliers such as interest rates, equity prices, or inflation or it is simply fixed at the start of the contract. There are many different variants of swaps which allow parties to swap the payments that they are receiving or making with another party based on different underlying cash flows.

Before we look at the different types of swaps, it is important to understand two key terms: floating and fixed payments.

- **Fixed** applies to payments that will not fluctuate over the payment period. These payments are known at the start, for example receiving 5% interest per year for the next 2 years. You can calculate exactly how much you need to pay in the future. The fixed rate may also be quotes as a spread above government bond yields, e.g. T + 35/40 basis points (where T is the prevailing yield for the current benchmark Treasury of the same maturity).
- **Floating** applies to payments that can fluctuate during the payment period. For example this may include paying interest calculated as the current LIBOR rate (i.e. rate at which banks lend to each other). As you can imagine, the amount to be paid each month differs so that there is uncertainty in the future payments.

A swaps contract can include any combination of these two payment specifications depending on the needs of the two parties involved. The parties involved in these types of transactions typically include:

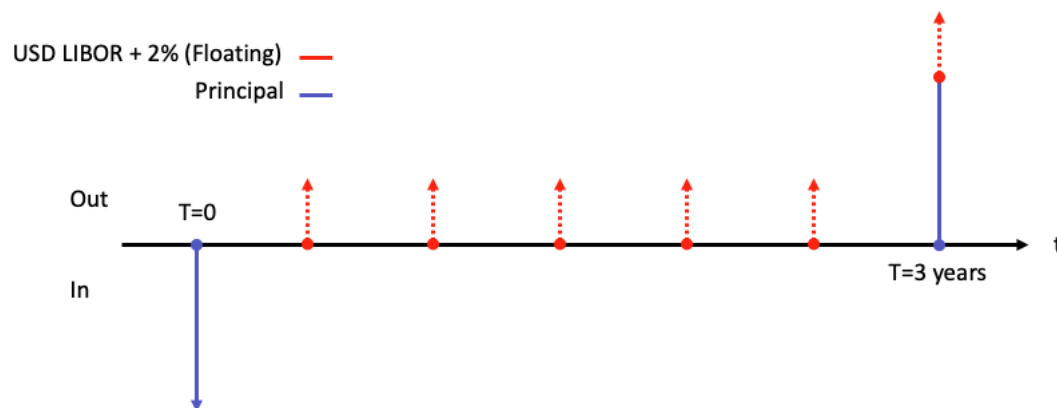
- | | |
|---------------------------|-----------------------|
| - Banks | - Hedge Funds |
| - Corporations | - Government Entities |
| - Institutional Investors | |

The following example will give a basic idea of how these identities may use swaps to hedge risk or speculate.

Example 5 - Interest Rate Swaps

*Consider that you are CEO of a large company and recently issued **£10 million in short term bonds with an expiry in 3 years with a variable coupon rate paid semi-annually** and calculated as the current London Interbank Offered Rate plus 2%. As you can imagine in the current economic climate, the future of interest rate rises is highly unpredictable and therefore you may be liable to pay larger amounts of interest (coupons) on the bonds if interest rates rise more than expected. To hedge against this risk and lock in the interest paid, you would enter a swap.*

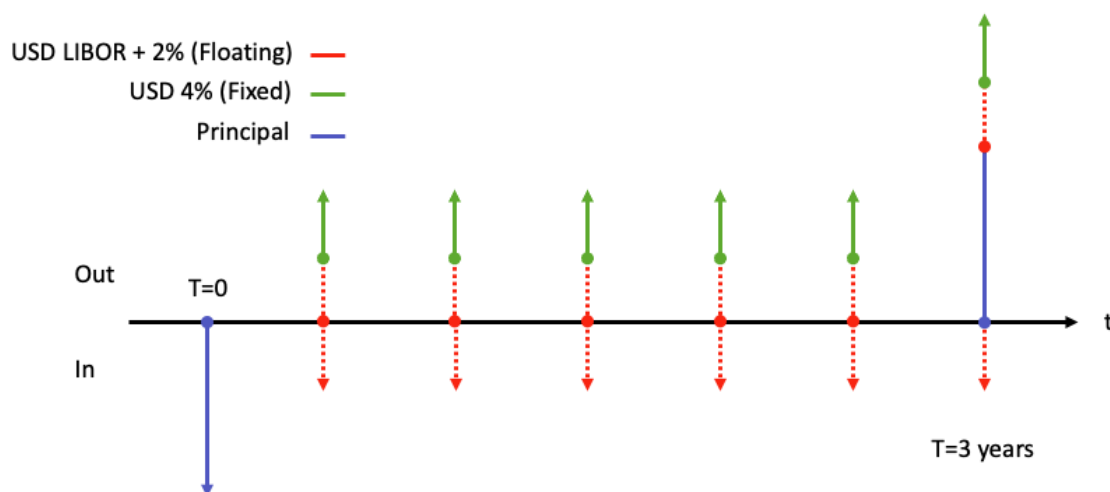
The diagram below shows your future payments before getting a swap.



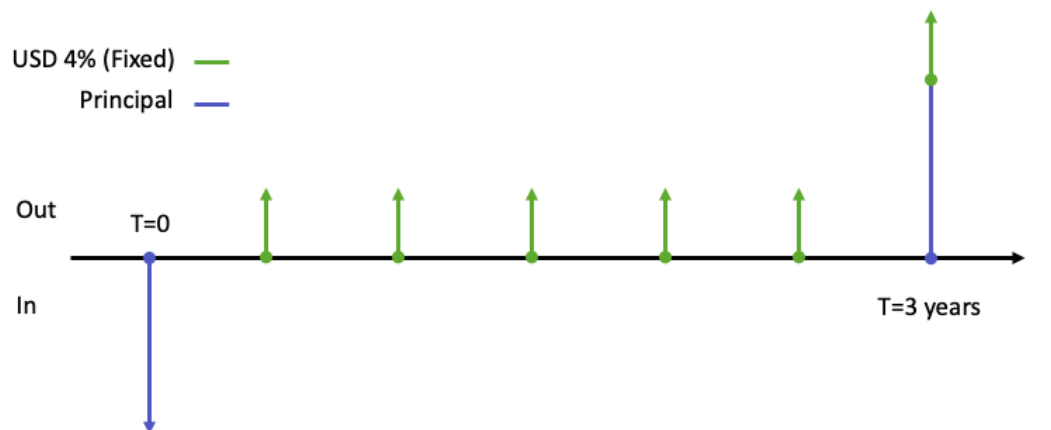
The dotted lines here represent the cash flows from the company paying investors coupons of an amount determined at the date that the coupon is to be paid - the floating rate. As the coupons are paid semi-annually, the **interest paid at each arrow is calculated as: Notional Principal $\times$ Interest Rate $\times$ Day Count Fraction**. The Day Count Fraction in this case is equal to 180/360 (semi-annual - US convention). Note also the principal payments at the start, receiving investor funds, and at the end, paying back bond principal at expiry.

To hedge the floating rate risk, the company enters a fixed-for-floating swap so that they can guarantee their future liabilities and pay a fixed rate. **You find a bank who is willing to pay you LIBOR + 2% on £10 million notional and in exchange you pay a fixed rate of 4% for the next three years to the bank.**

Now each month you still pay the LIBOR + 2% to investors but immediately receive LIBOR + 2% from the swaps contract you entered. The two cancel out. You are still left to pay the 4% fixed fee to the bank so overall you are not at risk if LIBOR increases. The new payment structure can be seen below.



As you can see, the two floating rates cancel to give simply a 4% fixed payment removing any risk of interest rates on the future payments on the bond.



You now make a 'profit' (compared to an unhedged position) if the floating payment exceeds 4% as you locked in a rate of 4% through the swaps contract.

There are a number of different variations on the mechanic described above including fixed-for-fixed payments, floating-for-floating payments (basis swaps) and even fixed-for-floating payments. These core mechanics can be applied to a wide range of underliers with the main contracts being:

Interest Rate Swaps - the size of the cash flows are linked to the interest rates in the same country. Here the floating payments are usually linked to a market standard rate like LIBOR or SOFR.

Currency Swaps - the size of the cash flows are linked to interest rates in different currencies

Asset Swaps - the fixed coupon of a specific bond is exchanged with a benchmark floating rate plus a margin

Inflation Swaps - the cash flows differ depending on the realised inflation in a particular country

To understand how a company may profit from a swap we can look at an example. Imagine you start a company, Simperial Ltd. and need some funds to run the business. Finally, your company decides on issuing £1 million in five-year bonds with a variable annual interest rate of 2% plus the federal funds rate. The federal funds rate fluctuates but now sits at around 1.75%. This means that you receive £10 million in funds immediately but have to pay back a rate of 2% plus the fed funds rate each year to the bond holders.

To hedge the risk faced by the fed funds rate increasing, and therefore increasing your interest payments, you find a company ABC Inc. willing to enter into an interest rate swap contract with you. ABC Inc. also owe £10 million on a loan they have but the interest that they pay is fixed at 4.5% per year. ABC Inc. likes the idea of a lower interest rate and can handle more risk than your company so you agree to swap cash flows. ABC Inc. pays your interest of 2% plus the fed funds rate and you pay their interest rate of 4.5%. Both debts are still paid but you now pay a fixed 4.5% while ABC Inc. covers your repayments. Table 2 shows how this scenario would play out if the fed funds rate happened to increase by 0.5% per year.

	Fed. Funds Rate + 2%	Variable Interest Paid by ABC Inc. to Simperial Ltd.	4.5% Interest Paid by Simperial Ltd. to ABC Inc.	ABC's P/L vs 4.5% without swap	Simperial's P/L vs floating rate without swap
Year 1	3.75%	£375,000	£450,000	£75,000	-£75,000
Year 2	4.25%	£425,000	£450,000	£25,000	-£25,000
Year 3	4.75%	£475,000	£450,000	-£25,000	£25,000
Year 4	5.25%	£525,000	£450,000	-£75,000	£75,000
Year 5	5.75%	£575,000	£450,000	-£125,000	£125,000
Total				-£125,000	£125,000

Table 2: Interest rate swap example for increasing floating rate

As you can probably see, if the fed funds rate increases above 3%, Simperial Ltd. will start saving money on the money repaid monthly to bond holders. A swap may also be favourable as lenders may be more likely to lend if your current loan obligations have fixed rates rather than floating rates.

Finally, the swaps contract that is famous for providing major profits to those who foresaw the 2008 financial crisis: credit default swaps. A credit default swap (CDS) is an agreement by one party to pay the lost principal and interest of a loan to the CDS buyer if a borrower defaults on a loan. More simply, this contract acts like insurance on a debt obligation, e.g. a bond. You can enter this contract with a financial institution OTC who will pay you the entire value of a bond's principal and future coupon payments if it defaults. In return, you pay a fee each set number of days that the bond has not defaulted.

CDS contracts can be very profitable as Michael Burry found in 2008 when his fund Scion Asset Management entered these agreements against mortgage backed securities. Who would've predicted that the strongest market in the world, real estate, would collapse as the majority of homeowners couldn't pay back their interest repayments? Obviously the banks that Bury entered these contracts with couldn't imagine this and so his fund only paid a low interest rate in comparison to the huge amount of assets he was able to 'insure'. Eventually, when the housing market did crash, these contracts paid out and he secured over \$700 million in profit for his investors.

Classifying derivatives

Futures and forwards are called vanilla derivatives because the price, expiry and quantity of underlying assets are fixed as soon as the contract is entered. It may help to remember this by thinking of vanilla having an etymology for basic, implying that futures and forwards are the simplest contracts traded. This is not the case for all derivatives as we will see with exotic contracts where any one of these variables may change throughout the contracts lifetime before expiry.

We can split the available derivatives into several types; vanilla, simple exotics, and complex exotics. The first class of vanilla derivatives are characterised by **fixed terms** that do not vary throughout the life of the contract. The terms that outline a derivative include:

- The time period over which it acts (or expiry)

- The price at which it is delivered for
- The quantity and quality of the underlying asset traded

A rule of thumb to follow is that even if just one term of the traditional vanilla structure is broken, then the derivative becomes an exotic. Simple exotics typically have only one term altered, for example an options contract which can be activated over a range of dates rather than just one. For more than one of the terms altered, the derivatives contract is known as a complex.